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# Variability of Biomass and Harvest Index from Several Soybean Genotypes as Renewable Energy Source

Ayda Krisnawati[a], M. Muchlish Adie[a]*

[a]*Indonesian Legumes and Tuber Crops Research Institutes, Jl. Raya Kendalpayak KM 8, Malang 65101, Indonesia*

**Abstract**

Soybeans is a legume crop that has economic value on seed yield, straw and biomass. This study was aimed to identify the biomass and harvest index from several soybean genotypes. The research was conducted at Pasuruan (East Java, Indonesia), from January to March 2014, using 29 genotypes. The analysis variance elucidated that days to maturity, straw weight, and harvest index were significant, except the seed yield. Seed yield varied from 1 977.58 kg $ha^{-1}$ to 3 008.03 kg $ha^{-1}$ with an average of 2 464.98 kg $ha^{-1}$, straw weight between 3 033.98 kg $ha^{-1}$ to 5 329.58 kg $ha^{-1}$ (average of 4 069.26 kg $ha^{-1}$), and biomass production varied from 38.40 kg $ha^{-1}$ $d^{-1}$ to 63.64 kg $ha^{-1}$ $d^{-1}$ with an average of 51.81 kg $ha^{-1}$ $d^{-1}$. The best genotype as source of biomass for energy was G27 with the highest yield, highest straw weight, and also the highest productivity of straw weight $d^{-1}$ $ha^{-1}$.


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*Keywords:* Biomass; energy; harvest index; soybean

* Corresponding author. Tel.: +62 813 8002 4424; fax: +62 341 801 496.
*E-mail address:* my_ayda@yahoo.com

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**Nomenclature**

| | |
|---|---|
| **CV** | coefficient of variation |
| **HI** | harvest index |
| **d** | day |
| **asl** | above sea level |
| **1 t** = | $10^3$ kg |

## 1. Introduction

Soybean is one of strategic commodity in Indonesia, after rice and maize. In 2014, soybean harvested area target is $1.5 \times 10^6$ ha with a production target of $2.7 \times 10^6$ t. Soybean is a raw material for industry, especially for tempeh and tofu; while the oil is used as an energy source. Besides its seeds, soybean straw has the potential to be developed and used as an alternative source of bioenergy [1]

According to [2], biofuels are classified as first generation biofuels that are directly related to a biomass that is edible. Second generation biofuels that are defined as fuels produced from a wide array of different feedstock, and the third generation of biofuels that are related to algae biomass but could to a certain extent be linked to utilization of $CO_2$ as feedstock. Biomass is defined as the biodegradable fraction of products and waste from agriculture, including vegetable species (feedstock seeds as peanut, sunflower, palm, castor bean, soybean, jatropha etc) and animal fat (tallow), forestry and related industries, as well as the biodegradable fraction of industrial and urban waste [3]. Globally, $140 \times 10^9$ t of biomass is generated every year from agriculture [4]. For example, based on the total production of straw in 2009 in China, the theoretical output of seven main crop straw is $820 \times 10^6$ t. Rice straw accounts for 25.0 % of the total amount; wheat straw is 18.3 %; maize straw is 32.3 %; soybean straw is 3.3 %; potato straw is 2.7 %; cotton straw is 3.2 % rape straw is 4.6 % [5].

Many biomass and agricultural derived materials have been suggested as alternative energy sources [6], due to its great contribution to the environment [7] and to its role as a strategic source of renewable energy in substitution to diesel oil and other petroleum-based fuels [8,9]. The use of biomass for energy also reduces dependency on the consumption of fossil fuel; hence, contributing to energy security and climate change mitigation [4].

Numerous researches and some technologies for energy production from biomass have been developed [10-12]. A report stated that agricultural biomass waste, which was equivalent to approximately $50 \times 10^9$ t of oil, converted to energy can substantially displace fossil fuel, reduce emissions of greenhouse gases and provide renewable energy to some $1.6 \times 10^9$ people in developing countries, which still lack access to electricity [4]. Another report stated that total amount of biomass usable for energy purposes is more than $3.7 \times 10^6$ t. Calculating three kg of biomass for one kg of oil of primary energy, the equivalent is $1.23 \times 10^6$ t [13]. Plants generating heat and electricity from straw have been operating in Denmark since 1996. The plant capacity is 10 MW of electricity and 23 MW of heat and it consumes 40 000 t of straw annually [14].

Utilization of soybean biomass as energy resource have been carried out used straw of five soybean cultivars, namely ‘Ika’, ‘Neoplanta’, ‘Tisa’, ‘Podravka’ and ‘Vita’ as energy resources in biofuel production. The average value for the biomass ranged from 2 571.10 kg ha$^{-1}$ to 3 374.70 kg ha$^{-1}$ with energy value 39 875.2 MJ ha$^{-1}$ to 48 580.0 MJ ha$^{-1}$, and equivalent with natural gas of 893.54 m$^3$ ha$^{-1}$ to 1 177.89 m$^3$ ha$^{-1}$ [1]. The objective of this study was to identify the soybean biomass and harvest index from several soybean genotypes.

## 2. Materials and methods

Genetic material evaluated consist of 29 soybean genotypes. The research was conducted at Pasuruan (East Java, Indonesia), from January to March 2014. The climate of Pasuruan based on Oldeman classification system is type E (< three wet month consecutively and > six dry month respectively), with altitude of 124 m asl. The experimental design was Randomized complete design with four replication. Each genotype was planted on 2.8 m × 4.5 m with 40 cm × 15 cm plant distance, two plants per hill. Fertilizer with a dose of 50 kg ha$^{-1}$ Urea, 100 kg ha$^{-1}$ SP36 and

75 kg ha$^{-1}$ KCl were applied before sowing time. Weed, insect and disease were controlled intensively. Observations included days of flowering (d), plant height (cm), seed yield (kg ha$^{-1}$), straw weight (kg ha$^{-1}$), and soybean harvest index.

The similarity distance between 29 soybean genotypes were assessed using cluster analysis based on the scores of the most significant principal components of principal components analysis. Cluster analysis was calculated using average linkage method.

## 3. Result and discussion

The analysis variance showed that days to maturity, straw weight, and harvest index were significant, except the seed yield (Table 1). Significant effect of genotype indicates that there were differences of character from 29 soybean genotypes. CV values varied from 1.55 % to 17.76%.

Table 1. Analysis of variance for seed yield, straw weight, and harvest index.

| Character | Mean square | | | CV (%) |
|---|---|---|---|---|
| | Replication | Genotype | Error | |
| Days to maturity (d) | 2.885 057 ns | 116.423 645 ** | 1.480 296 | 1.55 |
| Seed yield (kg ha$^{-1}$) | 551 484.827 * | 273 416.300 ns | 191 689.72 | 17.76 |
| Straw weight (kg ha$^{-1}$) | 45 734.943 7 ** | 12 243.273 5 ** | 3 423.628 6 | 14.38 |
| Harvest Index (HI) | 0.000 617 24 ** | 0.001 485 84 ** | 0.000 444 62 | 2.46 |

Note: * = significant at 5 % probability level ($p < 0.05$); ** = significant at 1 % probability level ($p < 0.01$), ns = not significant, CV = coefficient of variation.

Days to maturity of 29 soybean genotypes ranged from 70 d to 88 d, seed yield varied from 1 977.58 kg ha$^{-1}$ to 3 008.03 kg ha$^{-1}$ with an average of 2 464.98 kg ha$^{-1}$, straw weight between 3 033.98 kg ha$^{-1}$ to 5 329.58 kg ha$^{-1}$ with an average of 4 069.26 kg ha$^{-1}$ (Table 2).

Soybean straw reflects the biomass production which has multifunctional as soil ameliorant, feed and bioenergy sources. From 29 soybean genotypes tested had a wide range straw weight with an average of 4 069.26 kg ha$^{-1}$. Soybean straw as a biomass source depending on the variety and environment [15]. G27 had the straw weight reached 5 329.58 kg ha$^{-1}$. Research by Kiss et al. [1], soybean straw weight ranged from 2 571.10 kg ha$^{-1}$ to 3 374.70 kg ha$^{-1}$. Another research obtained the potential straw weight of soybean variety Anjasmoro higher than the varieties Tanggamus and Slamet [16]. Straw weight evaluation of 22 Indonesian soybean varieties in Bogor, Probolinggo and Banyuwangi on 2007 obtained straw weight between 1 307 kg ha$^{-1}$ to 3544 kg ha$^{-1}$ (In a conversation with a Ir. Abdullah Taufiq, MS from ILETRI, in January 2014).

Table 2. Days to maturity, seed and straw yield from 29 soybean genotypes.

| No | Genotype | Mean Value | | |
|---|---|---|---|---|
| | | Days to maturity (d) | Seed yield (kg ha$^{-1}$) | Straw weight (kg ha$^{-1}$) |
| G1 | A PSJ | 78 fg | 2 074.79 | 3 450.15 fghi |
| G2 | B PSJ | 78 g | 2 458.09 | 4 512.88 abcde |
| G3 | E PSJ | 72 jk | 2 210.89 | 4 197.00 bcdefg |
| G4 | F PSJ | 80 e | 2 138.68 | 4 147.33 bcdefg |
| G5 | G PSJ | 79 efg | 1 977.58 | 3 868.33 cdefghi |
| G6 | H PSJ | 83 cd | 2 302.55 | 4 748.23 abcd |
| G7 | G 511 H x Anjs-6-5 | 83 d | 2 319.21 | 4 453.05 abcdef |
| G8 | G 511 H x Anjs-6-6 | 83 d | 2 619.18 | 4 227.25 bcdefg |

Continued on next page

Table 2. Continued

| No | Genotype | Mean Value | | |
|---|---|---|---|---|
| | | Days to maturity (d) | Seed yield (kg ha$^{-1}$) | Straw weight (kg ha$^{-1}$) |
| G9 | G 511 H x Anjs-6-9 | 83 d | 2 666.40 | 4 208.55 bcdefg |
| G10 | G 511 H x Anjs-6-10 | 83 d | 2 510.86 | 4 211.80 bcdefg |
| G11 | G 511 H x Anjs-4-2 | 83 d | 2 519.19 | 4 75.06 abcd |
| G12 | G 511 H x Anjs-10-7 | 83 cd | 2 408.09 | 4 382.65 abcdef |
| G13 | Bio-bal-420 | 85 b | 2 765.00 | 4 204.60 bcdefg |
| G14 | Bio-bal-456 | 80 ef | 2 580.30 | 3 930.05 cdefghi |
| G15 | Bio-G-366 | 79 efg | 2 419.20 | 3 034.98 i |
| G16 | Bio-G-422 | 70 l | 2 094.24 | 3 143.45 hi |
| G17 | Bio-497 | 74 hi | 2 426.15 | 3 332.10 ghi |
| G18 | AnjsxArg-200-22 | 72 jk | 2 667.79 | 3 752.90 defghi |
| G19 | AnjsxArg-189-19 | 75 h | 2 428.93 | 3 516.50 efghi |
| G20 | AnjsxArg-169-15 | 74 hi | 2 566.41 | 3 877.03 cdefghi |
| G21 | AnjsxArg-235-27 | 71 kl | 2 574.74 | 4 092.55 bcdefgh |
| G22 | AnjsxMal-134-9 | 73 ij | 2 758.06 | 4 199.68 bcdefg |
| G23 | SinbgxMal-461-45 | 72 jk | 2 813.61 | 3 647.53 efghi |
| G24 | SinbgxMal-479-49 | 73 ij | 2 894.16 | 3 772.50 defghi |
| G25 | Grobogan | 70 l | 2 463.65 | 3 477.45 fghi |
| G26 | Baluran | 85 b | 2 024.80 | 3 731.63 efghi |
| G27 | Dering-1 | 84 cbd | 3 008.03 | 5 329.58 a |
| G28 | Panderman | 88 a | 2 266.44 | 4 965.48 ab |
| G29 | Anjasmoro | 85 cb | 2 527.53 | 4 843.95 abc |
| Average | | 78 | 2 464.98 | 4 069.23 |

Mean followed by the same letter in the same column are not significantly different at 5 % level according to DMRT test

Various researches showed that the use of soybean straw as source of bioenergy were encouraging. A study was conducted to evaluate the potential use of the soybean straw for the production of liquid fuel intermediates. The immediate goal was to demonstrate production of pyrolysis liquid that can be burned potentially and be upgraded to transportation grade fuel and at the same time produce biochar that can be deployed as a soil amendment. The results shows that high yields of pyrolysis liquids (bio-oil) can be efficiently produced from the soybean straws [11]. Another research concludes that soybean straw is a good basis for the production of the second generation of biofuels from renewable resources by biomass-to-liquid procedure [1].

As discussed earlier, soybean has potential in producing seed and straw. Straw ratio and harvest index from 29 genotypes was presented in Table 3. Harvest index is a ratio between economic character (seed yield) and total plant dry weight (seed yield + straw weight) which is showed the proportion of seed yield and straw weight.

The harvest index average of 29 soybean genotypes was 0.38, showed that a total of 38 % was seed yield and the rest (72 %) was straw yield. Genotype G27 had the highest seed yield (3 008.03 kg ha$^{-1}$), and also the highest straw weight (5 329.58 kg ha$^{-1}$) with harvest index value of 0.36.

Table 3. Straw ratio and harvest index from 29 soybean genotypes.

| No | Genotype | Mean Value | |
|---|---|---|---|
| | | Harvest Index (HI) | Straw weight $d^{-1}$ (kg $ha^{-1}$) |
| G1 | A PSJ | 0.38 abcdef | 44.09 |
| G2 | B PSJ | 0.35 cdefg | 58.04 |
| G3 | E PSJ | 0.34 defg | 58.29 |
| G4 | F PSJ | 0.34 defg | 51.68 |
| G5 | G PSJ | 0.34 fg | 49.28 |
| G6 | H PSJ | 0.33 efg | 57.21 |
| G7 | G 511 H x Anjs-6-5 | 0.34 defg | 53.98 |
| G8 | G 511 H x Anjs-6-6 | 0.38 abcdef | 51.08 |
| G9 | G 511 H x Anjs-6-9 | 0.39 abcdef | 51.01 |
| G10 | G 511 H x Anjs-6-10 | 0.37 abcdef | 50.90 |
| G11 | G 511 H x Anjs-4-2 | 0.35 cdefg | 57.41 |
| G12 | G 511 H x Anjs-10-7 | 0.35 cdefg | 52.80 |
| G13 | Bio-bal-420 | 0.40 abcd | 49.47 |
| G14 | Bio-bal-456 | 0.39 abcde | 49.13 |
| G15 | Bio-G-366 | 0.44 a | 38.40 |
| G16 | Bio-G-422 | 0.40 abcd | 44.91 |
| G17 | Bio-497 | 0.42 ab | 45.03 |
| G18 | AnjsxArg-200-22 | 0.42 abc | 52.12 |
| G19 | AnjsxArg-189-19 | 0.41 abcd | 46.89 |
| G20 | AnjsxArg-169-15 | 0.40 abcd | 52.39 |
| G21 | AnjsxArg-235-27 | 0.39 abcde | 57.64 |
| G22 | AnjsxMal-134-9 | 0.40 abcd | 57.53 |
| G23 | SinbgxMal-461-45 | 0.44 ab | 50.66 |
| G24 | SinbgxMal-479-49 | 0.43 ab | 51.68 |
| G25 | Grobogan | 0.42 abc | 49.68 |
| G26 | Baluran | 0.35 defg | 43.90 |
| G27 | Dering-1 | 0.36 bcdefg | 63.64 |
| G28 | Panderman | 0.31 g | 56.43 |
| G29 | Anjasmoro | 0.34 defg | 57.16 |
| Average | | 0.38 | 51.81 |

Mean followed by the same letter in the same column are not significantly different at 5 % level according to DMRT test

Selection of soybean genotypes that have the genetic potential for producing biomass and also the identification of soybean genotypes that producing both of high yield potential and high biomass was performed by cluster analysis. Clustering of 29 soybean genotypes based on straw weight was presented in the Fig. 1 and Table 4. The grouping classified the straw weight of 29 genotypes into five categories from very high to very low weight. Genotype G27 had the highest straw weight. Genotype G6, G11, G29, and G28 with straw weight range between 4 783.23 kg $ha^{-1}$ to 4 956.48 kg $ha^{-1}$ and categorized as genotypes with high straw weight.

Table 4. Soybean clustering based on straw yield (biomass) of 29 soybean genotypes.

| Cluster | Genotypes | Straw weight | Category |
|---|---|---|---|
| I | 6, 11, 29, 28 | 4 783.23 to 4 965.48 | High |
| II | 3, 22, 9, 10, 13, 8, 4, 21, 2, 7, 12 | 4 197.00 to 4 382.65 | Medium |
| III | 5, 20, 18, 24, 1, 25, 26, 19, 14, 23, 17 | 3 332.10 to 3 868.33 | Low |
| IV | 15, 16 | 3 033.98 to 3 143.45 | Very low |
| V | 27 | 5 329.58 | Very high |

Note: Genotypes code refer to Table 2.

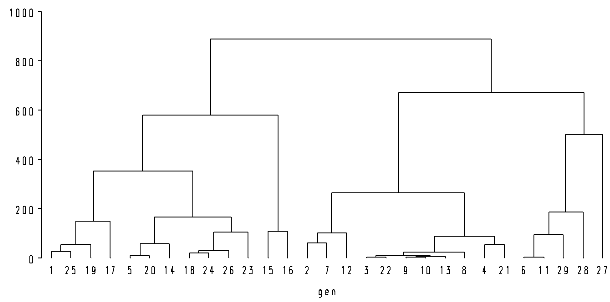


Fig. 1. Dendogram derived from Cluster based on straw weight of soybean genotypes.

Grouping as in Figure 1, only illustrates the potential straw weight, and did not consider the potential seed yield. Figure 2 and Table 5 classify 29 soybean genotypes based on the harvest index, which divide the 29 soybean genotypes into five clusters. There were six genotypes of soybean with harvest index from 0.42 to 0.43 (classified as very high), the seed yields between 42 % to 43 % and the potential biomass from 57 % to 58 %. These six have both of high genetic potential yield and high biomass.

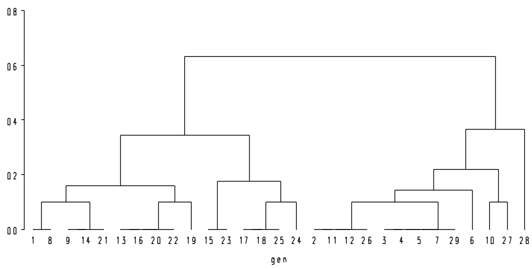


Fig. 2. Dendogram derived from Cluster based on harvest index of soybean genotypes.

Table 5. Soybean clustering based on harvest index of 29 genotypes.

| Cluster | Genotypes | Harvest index | Category |
|---|---|---|---|
| I | 3, 4, 5, 7, 2, 11, 12, 26, 29, 6 | 0.33 to 0.34 | Low |
| II | 1, 8, 9, 14, 13, 16, 20, 21, 22, 19 | 0.38 to 0.41 | High |
| III | 17, 18, 15, 23, 25, 24 | 0.42 to 0.43 | Very high |
| IV | 10, 27 | 0.36 to 0.37 | Medium |
| V | 28 | 0.34 | Very low |

Note: Genotypes code refer to Table 2.

Utilization soybean biomass as source of bioenergy in Indonesia is very prospective. This is due to the high of harvest index ratio and supported by early maturing soybean, which is only 78 d. Biomass production of 29 genotypes was varied, from 38.40 kg $ha^{-1}$ $day^{-1}$ to 63.64 kg $ha^{-1}$ $day^{-1}$ with an average of 51.81 kg $ha^{-1}$ $d^{-1}$. Interesting characteristic comes from genotype G27, which had the highest seed yield (3 008.03 kg $ha^{-1}$), the highest straw weight (5 329.58 kg $ha^{-1}$), and also producing the highest productivity of straw weight $ha^{-1}$ $d^{-1}$ (63.64 kg $ha^{-1}$ $d^{-1}$).

## 4. Conclusion

Genotype G27 had the highest yield (3 008.03 kg $ha^{-1}$), highest straw weight (5 329.58 kg $ha^{-1}$), and also the highest productivity of straw weight $ha^{-1}$ $d^{-1}$ (63.64 kg $ha^{-1}$ $d^{-1}$), hence potential to be developed as soybean biomass for bioenergy source.

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## References

[1] Kis D, Sucic B, Guberac V, Voca N, Rozman V, Sumanovac L. Soybean biomass as a renewable energy resources. Agriculture Conspectus Scientificus 2009; 74 (3): 201-203.
[2] Lee RA, Lavoie JM. From first- to third-generation biofuels: Challenges of producing a commodity from a biomass of increasing complexity. Animal Frontier 2013; 3 (2): 11p.
[3] Da Silva ST, Dutra C. Brazilian policy of biodiesel: alternative to mitigate effects of climate change? Brazil: UNISANTOS; 2008.
[4] UNEP (United Nations Environmental Programme). Converting waste agricultural biomass into a resource. Compendium of Technology. United Nations Environment Programme; 2009: p.441.
[5] Tian Y, Sun L. Resource, policy and status and prospect on agricultural bio-energy in China. 2nd Regional bioenergy policy dialogue: identifying priority areas for regional cooperation and action. Chinese Academy of Agricultural Engineering, MOA; 2011. p.40.
[6] Mushrush G, Beal EJ, Spencer G, et al. An environmentally benign soybean derived fuel as a blending stock or replacement for home heating oil. J. Environ. Sci. Health 2001; 36: 613-622.
[7] Sergej U. Biomass as a renewable source of energy in the Czech Republic. Proceedings of the Workshop "Cereals straw and agricultural residues for bioenergy in European Union New Member States and Candidate Countries". In Scarlat N, Dallemand JF, Martinov M, editors, Novi Sad, Serbia. Italy: Joint Research Centre Institute for Environment and Sustainability; 2008.
[8] Bagley ST, Gratz JF, Jonson LD, Mcdonald JH. Effects of an oxidation catalytic converter and a biodiesel fuel on the chemical, mutagenic and particle size characteristics of emissions from a diesel engine. Environ. Sci. Technol. 1998; 32: 1183-1191.
[9] Wu WH, Foglia TA. Property and engine performance evaluation of ethyl and isopropyl esters of tallow and grease. J. AOCS 1998; 75: 1173-1178.
[10] Cardone M, Prati MV, Rocco V, Seggiani M, Senatore A, Vitolo S. Brassica carinata as an alternative oil crop for the production of biodiesel in Italy: engine performance and regulated and unregulated exhaust emissions. Environ. Sci. Technol. 2002; 36: 4656-4662.
[11] Mladenović R, Paprika M, Komatina M, Erić A, Mladenović M, Dakić D. Boiler combusting large soya straw bales for heating the greenhouses complex. FME Transactions 2008; 36: 27-32.
[12] Boateng AA, Mullen CA, Goldberg NM, et al. Sustainable production of bioenergy and biochar from the straw of high-biomass soybean lines via fast pyrolysis. Environmental Progress & Sustainable Energy 2010; 29 (2): 175–183.
[13] Martinov M, Djatkov D, Golub M, Viskovic M. Harvestable biomass of corn, wheat, soybean and sunflower, experiences in Vojvodina (agricultural region of Serbia). Joint Research Centre of the European Commission in collaboration with the Energy Institute Hrvoje Požar. Enlargement & Integration Programme Expert Consultation. "The scientific basis for sustainable use of biomass residues and wastes for bioenergy and biofuels". Zagreb, Croatia; 2013.
[14] Forgie V, Andrew R. Life cycle assessment of using straw to produce industrial energy in New Zealand. Landcare Research and New Zealand Centre for Ecological Economics 2009; p.37.
[15] Martinov M, Tesic. Cereal/soybean straw and other crop residues utilization as fin Serbia–status and prospects. Proceedings of the Workshop "Cereals straw and agricultural residues for bioenergy in European Union New Member States and Candidate Countries". In Scarlat N, Dallemand JF, Martinov M, editors, Novi Sad, Serbia. Italy: Joint Research Centre Institute for Environment and Sustainability; 2008.

[16] Sari I. Studi ketersediaan dan serapan hara mikro serta hasil beberapa varietas kedelai pada tanah gambut yang diameliorasi abu janjang kelapa sawit [Study of micronutrient availability and uptake and soybean varieties results on peat soil applicated oil palm bunch ash]. J. Enviagro Pertanian dan Lingkungan 2011; 4 (1): 1-48 [Bahasa Indonesia].